AQUAMIND

Water Tank Control Mechanism

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Digital Logic Design (CSC 120)

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INTRODUCTION

Water is essential in our daily lives, and because of this constant need, water tanks have become common on the rooftops of most homes nowadays. However, traditional water tanks come with several challenges. These challenges motivated us to make our project (Water Tank Control Mechanism), which aims to solve issues such as:

• Monitoring the water level inside the tank
• Automatically refilling and draining the tank without human intervention
• Providing an alert system to prevent overflow

In this project, we will take you through our journey of applying the knowledge we've gained so far to solve these problems most efficiently and practically.

WATER LEVEL DISPLAY FUNCTION.

This part of our project aims to display the water level in the tank using a 7-segment display.
First , we take readings from 9 sensors in the water tank, each one located after 10% as an input, then we convert them to 4-bits BCD.

Truth Table 1

a	b	c	d	e	f	g	h	i	Decimal no.	A	B	C	D
0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	1	0	0	0	1
1	1	0	0	0	0	0	0	0	2	0	0	1	0
1	1	1	0	0	0	0	0	0	3	0	0	1	1
1	1	1	1	0	0	0	0	0	4	0	1	0	0
1	1	1	1	1	0	0	0	0	5	0	1	0	1
1	1	1	1	1	1	0	0	0	6	0	1	1	0
1	1	1	1	1	1	1	0	0	7	0	1	1	1
1	1	1	1	1	1	1	1	0	8	1	0	0	0
1	1	1	1	1	1	1	1	1	9	1	0	0	1

Boolean Equations from Truth Table (1)

We used SOP (sum of product) to find the equations from the truth table

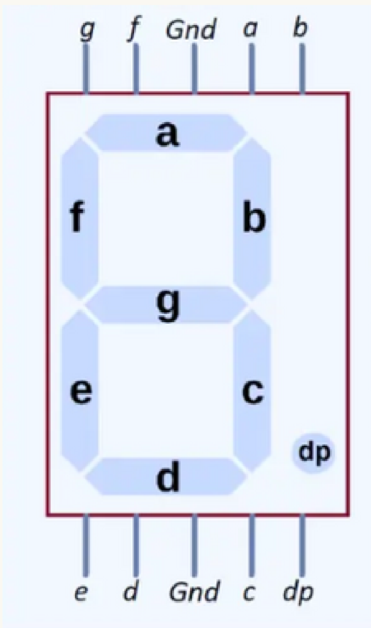
1) $A = abcdefghi' + abcdefghi$
2) $B = abcde'f'g'h'i' + abcdef'g'h'i' + abcdefg'h'i' + abcdefgh'i'$
3) $C = abc'd'e'f'g'h'i' + abcd'e'f'g'h'i' + abcdefg'h'i' + abcdefgh'i'$
4) $D = ab'c'd'e'f'g'h'i' + abcd'e'f'g'h'i' + abcdef'g'h'i' + abcdefgh'i' + abcdefghi$

Simplified equations

We used Boolean algebric rules to simplify the expressions to:

1) $A = abcdefgh$
2) $B = abcdf'g'h'i' + abcdefh'i'$
3) $C = abd'e'f'g'h'i' + abcdefh'i'$
4) $D = ad'e'f'g'h'i'(b \text{ XNOR } c) + abcdef'g'h'i' + abcdefgh'i' + abcdefghi$

The Binary Coded Decimal (BCD) output from the first truth table (consisting of 4 bits) is used as the input to a second truth table. This second truth table generates 7 output signals, labeled from 'a' to 'g'. These outputs correspond to the 7 segments of a 7-segment display. Each segment is an individual LED that can be turned on or off. By lighting up specific combinations of these segments, the display forms digits from 0 to 9 to display the level of water in the tank



Truth Table 2

A	B	C	D	Decimal no.	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	1	0	1	1	0	0	0	0
0	0	1	0	2	1	1	0	1	1	0	1
0	0	1	1	3	1	1	1	1	0	0	1
0	1	0	0	4	0	1	1	0	0	1	1
0	1	0	1	5	1	0	1	1	0	1	1
0	1	1	0	6	1	0	1	1	1	1	1
0	1	1	1	7	1	1	1	0	0	0	0
1	0	0	0	8	1	1	1	1	1	1	1
1	0	0	1	9	1	1	1	1	0	1	1

Boolean Equations from Truth Table (2)

We used Karnaugh map (K- Map), to find the equations from the truth table

1) a = B’D’ + C + DB +A

C

1	0	1	1
0	1	1	1
Φ	Φ	Φ	Φ
1	1	Φ	Φ

B

A

D

2) b = B’ + C’D’ + CD

C

1	1	1	1
1	0	1	0
Φ	Φ	Φ	Φ
1	1	Φ	Φ

B

A

D

Boolean Equations from Truth Table (2)

3) c = C′ + B + D

			C	C	
	1	1	1	0	
	1	1	1	1	B
A	Φ	Φ	Φ	Φ	B
A	1	1	Φ	Φ	
		D	D		

4) d = A + B′D′ + B′C + CD′ + BC′D

			C	C	
	1	0	1	1	
	0	1	0	1	B
A	Φ	Φ	Φ	Φ	B
A	1	1	Φ	Φ	
		D	D		

Boolean Equations from Truth Table (2)

5) e = B’D’ + CD’

C

1	0	0	1
0	0	0	1
Φ	Φ	Φ	Φ
1	0	Φ	Φ

B

A

D

6) f = A + C’D’ + BC’ + BD’

C

1	0	0	0
1	1	0	1
Φ	Φ	Φ	Φ
1	1	Φ	Φ

B

A

D

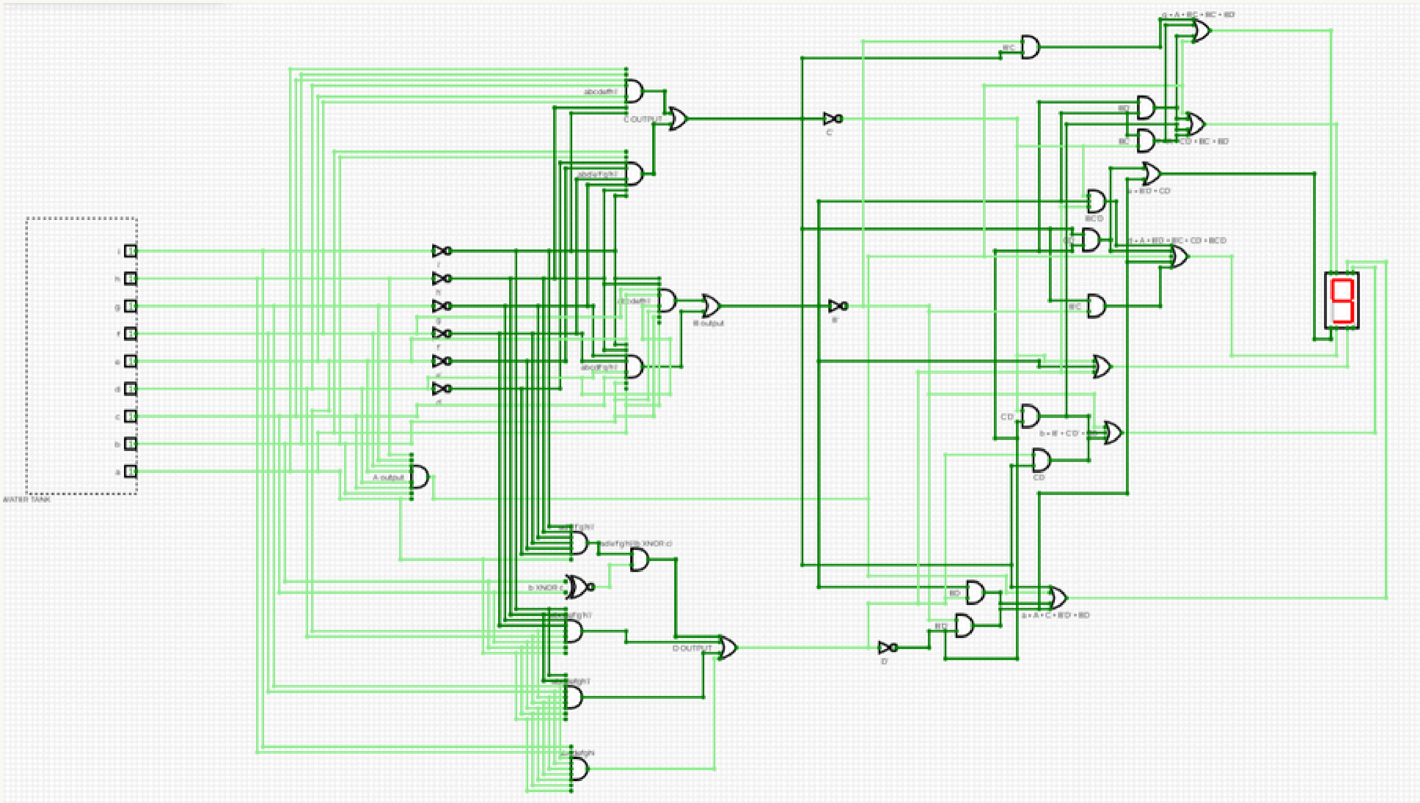
Boolean Equations from Truth Table (2)

7) $g = A + B'C + BC' + BD'$

			C		
	0	0	1	1	
	1	1	0	1	B
A	Φ	Φ	Φ	Φ	
	1	1	Φ	Φ	
		D			

Circuit with water level display

After deriving all the Boolean equations for the water level display function, we implemented them in CircuitVerse using basic logic gates (AND, OR, NOT, XNOR) to test and verify their functionality.



As shown in the image above, the 7-segment display is indicating the number 9. This is because all nine sensors in the tank are activated (each having a value of 1), which means the water has reached and covered all the sensors. This indicates that the tank is 90% full.

AUTO REFILL AND DRAIN CONTROL FUNCTION

This part of our project manages the automatic refilling and draining of the water tank. When the water level is low, the system activates the inlet valve to refill the tank; if the tank is at risk of overflowing, the outlet valve is triggered to release excess water. The decision-making process is based on the readings from nine water level sensors, which provide real-time input to determine whether to add or reduce water.

Truth Table 3

a	b	c	d	e	f	g	h	i	State	Vi	Vo
0	0	0	0	0	0	0	0	0	0	1	0
1	0	0	0	0	0	0	0	0	1	1	0
1	1	0	0	0	0	0	0	0	2	1	0
1	1	1	0	0	0	0	0	0	3	1	0
1	1	1	1	0	0	0	0	0	4	1	0
1	1	1	1	1	0	0	0	0	5	1	0
1	1	1	1	1	1	0	0	0	6	1	0
1	1	1	1	1	1	1	0	0	7	1	0
1	1	1	1	1	1	1	1	0	8	1	0
1	1	1	1	1	1	1	1	1	9	0	1

Boolean Equations from Truth Table (3)

We used SOP (sum of product) to find the equation of the inlet valve (Vi) from the truth table:

1) $Vi = a' + b' + c' + d' + e' + f' + g' + h' + i'$

And we used POS (product of sum) to find the equation of the outlet valve (Vo) from the truth table:

2) $Vo = abcdefghi$

Simplified equations

We used Boolean algebric rules to simplify the expression of the inlet valve to:

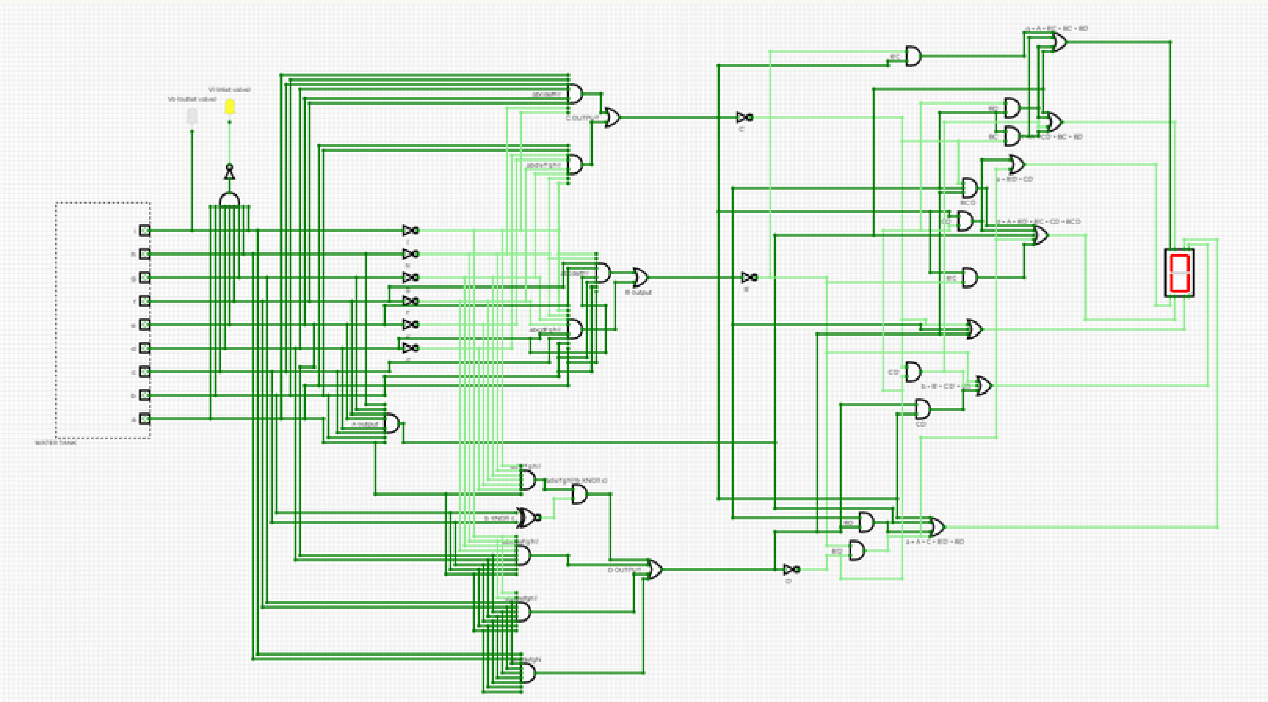
1) $Vi = (abcdefghi)'$

While the outlet valve expression can simply be written as:

2) $Vo = i$

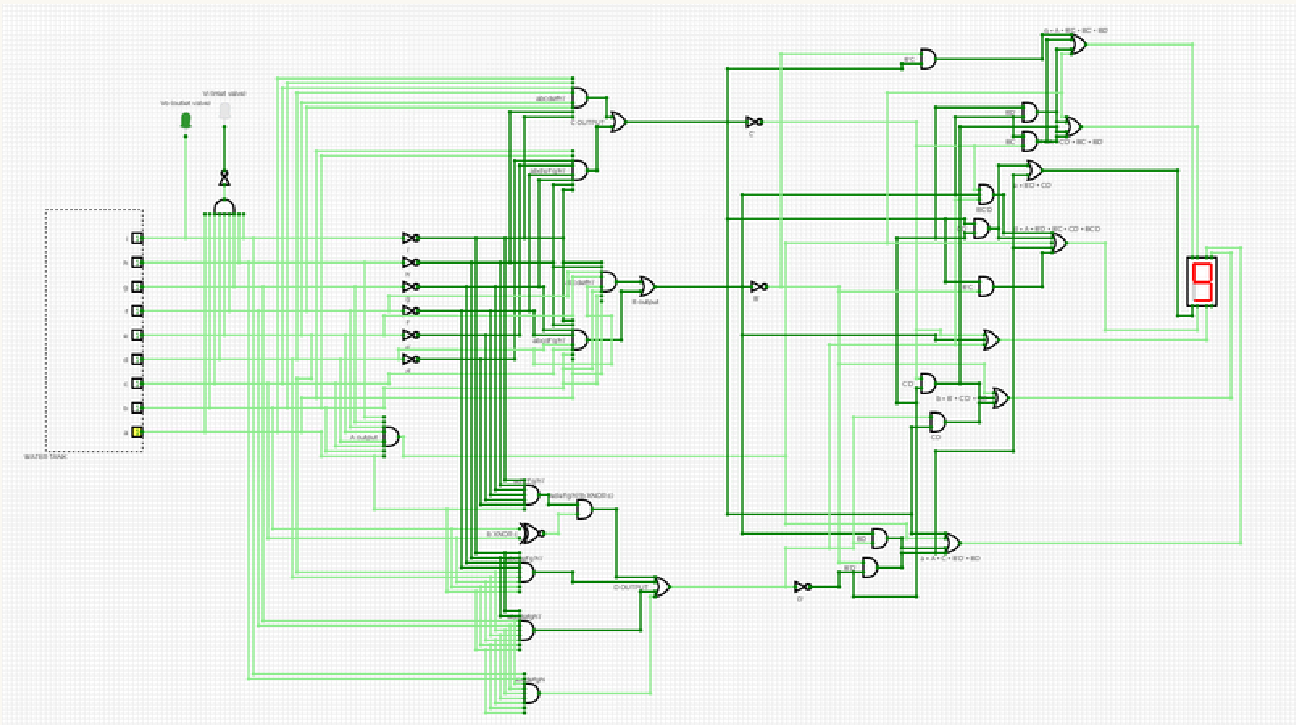
Circuit with Auto Refill and Drain Control

After deriving all the Boolean equations for the Auto Refill and Drain Control function, we implemented them in CircuitVerse using basic logic gates (AND, NOT). Since the simulator does not support valve components, we used LEDs to represent the valve states (yellow for the inlet valve and green for the outlet valve). This setup allowed us to test and verify the logic functionality effectively.



As shown in the image above, the inlet valve is open due to a very low water level. None of the nine sensors are activated, indicating that the tank is empty. Consequently, the 7-segment display shows the number 0.

Circuit with Auto Refill and Drain Control



In this image, the outlet valve is open because the water level is high all nine sensors are activated, indicating that the tank is full. This is also reflected on the 7-segment display, which shows the number 9.

WATER OVERFLOW ALARM FUNCTION

This part of our project is designed to provide an alert system that activates a buzzer when the water level in the tank approaches the last sensor, serving as an early warning mechanism.

Truth Table 4

a	b	c	d	e	f	g	h	i	State	Buzzer
0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	1	0
1	1	0	0	0	0	0	0	0	2	0
1	1	1	0	0	0	0	0	0	3	0
1	1	1	1	0	0	0	0	0	4	0
1	1	1	1	1	0	0	0	0	5	0
1	1	1	1	1	1	0	0	0	6	0
1	1	1	1	1	1	1	0	0	7	0
1	1	1	1	1	1	1	1	0	8	0
1	1	1	1	1	1	1	1	1	9	1

Boolean Equations from Truth Table (4)

We used SOP (sum of product) to find the equation of the buzzer from the truth table:

- Buzzer = abcdefghi

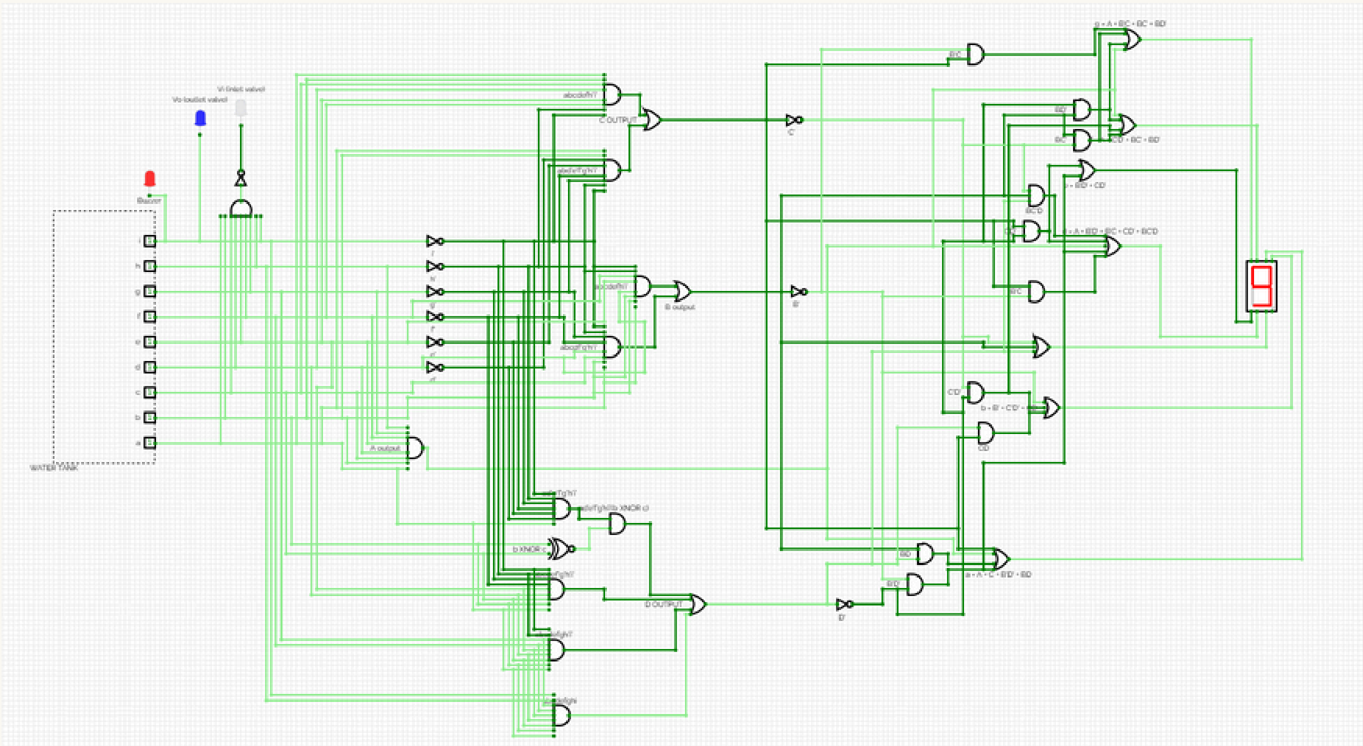
Simplified equations

The expression of the buzzer can simply be written as:

- Buzzer = i

Circuit with Water Overflow Alarm

After deriving all the Boolean equations for the water overflow alarm function, we implemented them in CircuitVerse. As the simulator does not support a buzzer component, a red LED was used to represent the buzzer state. This approach enabled us to effectively test and validate the logic functionality.



As shown in the image above, the buzzer is activated because the 9th sensor is triggered, indicating that the water has reached the highest level in the tank. This serves as a warning of a potential overflow risk.

HARDWARE IMPLEMENTATION

After completing the theoretical phase of the project (truth tables, Boolean expressions, k-maps, and circuit design) we decided to take the project a step further by implementing the core functionalities (water level display, overflow alarm, and automatic refill system) using more advanced techniques. This involved utilizing an Arduino microcontroller to bring our design into a practical, real-world application.

Hardware Components

To implement our project we used :

- Arduino UNO
- 4 Water sensors
- I2C LCD (16*2)
- Buzzer
- Solenoid valve
- Water pump
- 2 Relay module
- Breadboard
- Power supply (12V)
- Jumper wires
- 2 containers

Hardware Design

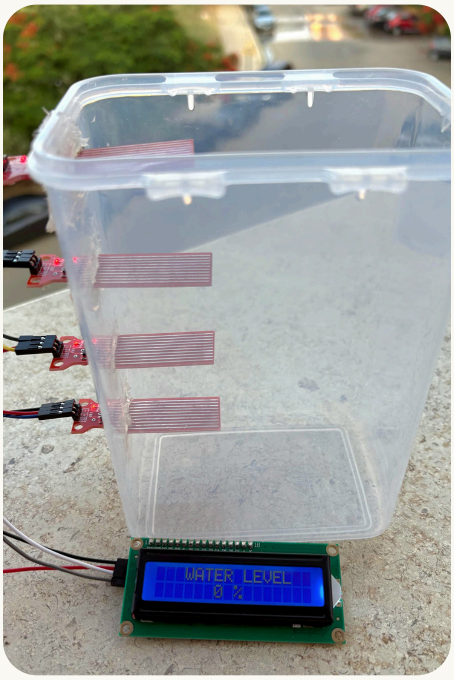
The hardware setup consists of two containers: the main water tank, which is equipped with four water level sensors, and a secondary supply tank that refills the main tank through a solenoid valve.

1. Water Level Sensing: Four sensors are placed inside the main tank at 25% intervals to monitor the water level accurately.
2. Display: Based on the sensor readings, an LCD screen displays the current water level as a percentage (25%, 50%, 75%, or 100%).
3. Refill Mechanism: When the water level falls below the maximum (one or more sensors are not triggered), the solenoid valve in the secondary tank opens automatically to refill the main tank.
4. Overflow Alert: When the main tank reaches full capacity (the fourth sensor is triggered and the LCD shows 100%), a buzzer is activated to alert for overflow.

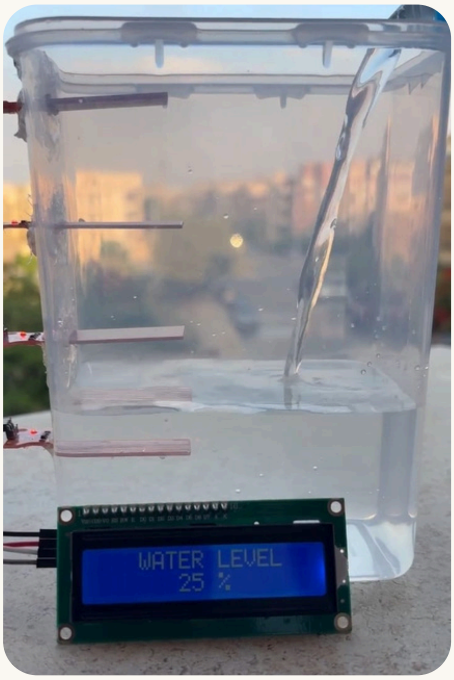
Visualizing the Water Level Indicator in Action

To bring our project to life, we've captured the real-time operation of the water level indicator at different stages of the tank's fill cycle. Each image below shows how the LCD display reflects the current water level as detected by the sensors:

Stage 1: Tank is completely empty — LCD shows 0%.

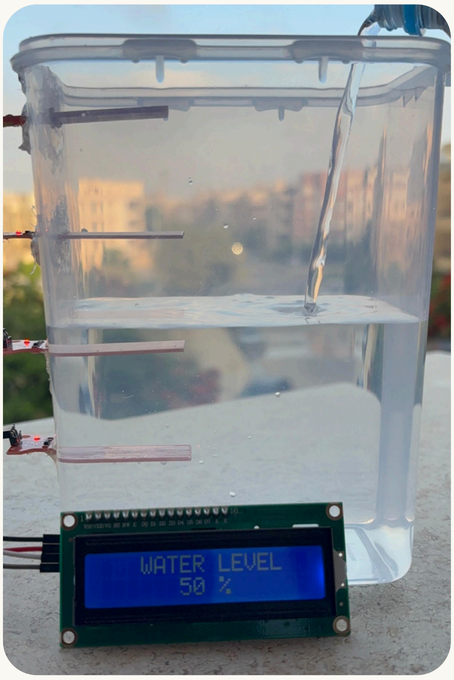


Stage 2: Water reaches the first sensor — LCD shows 25%.

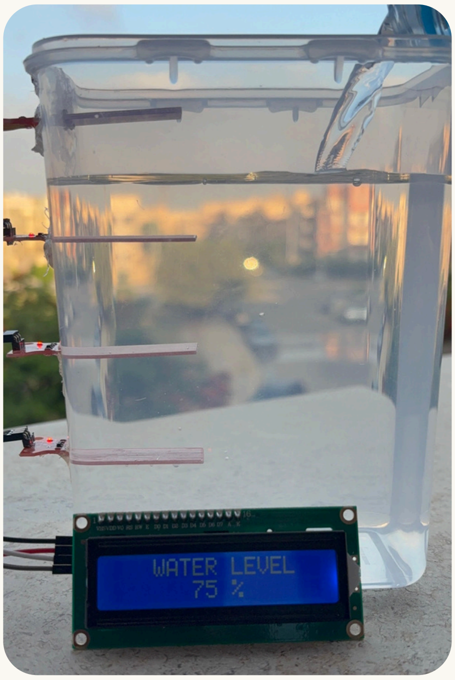


Visualizing the Water Level Indicator in Action

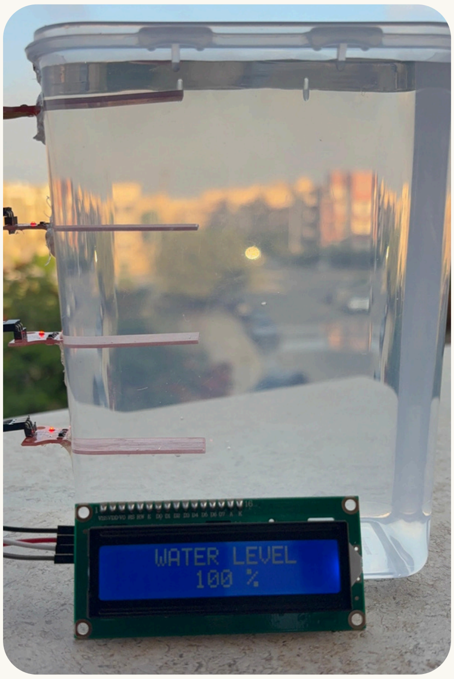
Stage 3: Water reaches the second sensor — LCD shows 50%.



Stage 4: Water reaches the third sensor — LCD shows 75%.



Stage 5: Water reaches the fourth sensor — LCD shows 100%.



CONCLUSION

This project demonstrated a successful integration of digital logic design with real-world hardware implementation to develop an efficient and automated Water Tank Control Mechanism. By systematically applying truth tables, Boolean algebra, and logic gate circuits, we translated theoretical concepts into a practical, functioning system.

The final design effectively monitors water levels, displays real-time status on an LCD, manages automatic refilling, and includes an overflow alert system. Each component was carefully tested and validated, both through simulation and physical hardware implementation using an Arduino based setup.

In conclusion, we all believe this project stands out as one of our most rewarding achievements, not only because it introduced a unique idea, but also because we thoroughly enjoyed collaborating on it. The experience gave us a strong sense of accomplishment, and we are proud of our performance, the success we achieved, and the valuable knowledge we gained throughout the process.